Phu Nguyen

For this lab I went through the notebook in order and ran the code myself so I could see what each part was doing. I trained polynomial regression models with degrees 1, 4, and 15 on the noisy sine wave data and plotted them together to compare. After that I used learning_curve to plot the training and cross-validation scores for the degree 1 and degree 15 models.

I also did the two tasks and the knowledge check questions. After that I ran the code, looked at the plots myself, and wrote the answers so they made sense to me.

This lab made me see how model complexity affects how well a model works on new data. When I finished, I uploaded it to my GitHub repo and submitted it.

Cyndy Ndong

I completed this entire lab assignment on my own from start to the end. I began by setting up my Google Colab environment and loading the dataset. I carefully inspected the data structure to understand what columns were present and checked for any missing values or weird formatting that needed to be cleaned up. After that, I separated the data set into input features and target labels so the model would know what to learn from.
Once the data was separated, I applied standard scaling to normalize all the numerical values. This step was important because it put all the features on an equal scale so large numbers wouldn't overpower smaller ones during training. After preparing the dataset, I selected and trained the classification model, making sure all parameters were set correctly. To evaluate the model properly, I wrote code to perform five-fold cross validation instead of relying on a simple train-test split. This allowed me to test the model across different subsets of the data and get a true average performance score.
I also calculated additional evaluation metrics including precision, recall, and accuracy scores to get a complete picture of how the model was performing. To make the results easy to read, I plotted a confusion matrix visualization that clearly showed where the model made accurate predictions and where it got confused.
Throughout the assignment, I ran into several common technical bugs like path errors, missing variable definitions, and column name mismatches. I systematically debugged each issue by inspecting my data frames, fixing line errors, and running cells in sequential order to keep the notebook clean. Finally, I reviewed the entire notebook, cleared unnecessary outputs, re-ran all code cells from top to bottom to guarantee reproducibility, and formatted all required submission documents.